Handoff — state of the project, 28 September 2026

Written for the next session (human or Claude) picking this up on another machine. It says what exists, what is verified, what is open and what to do next. `docs/status.md` remains the long-form record; this is the short one.

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1. Standing rules — these override defaults

1. **Safety and reliability come before cost, size and convenience.** Check at worst-case corners (datasheet min/max, −40 °C and +125 °C, 105 °C board ambient, reverse battery, ISO 7637-2 pulse 2a), never at typical. Prefer a topology that removes a failure mechanism over one that manages it. State failure modes and which direction is safe.
2. **No AI attribution in commits.** No `Co-Authored-By`, no "Generated with" line. (Commits `1332322`..`7a189fe` carry such trailers from before the rule; they are left alone deliberately — do not rewrite that history.)
3. **Deliverables are files in this repository.** No externally hosted pages.
4. **Verify with the project venv and `set -o pipefail`.** Never a bare `python3`, and never `| tail` in a chain that gates a commit — a piped tail once let a 22/24 suite into a commit claiming 24/24.
5. **Nothing is claimed without a check that runs.** A caveat in prose gets believed; a corner that is written as an enforced check is the one that gets caught. Every finding in this project was found this way.

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2. Setting up on a new machine

```
git clone https://github.com/mohan4220/ecu25kva.git
cd ecu25kva
python3 -m venv .venv                # .venv is git-ignored
.venv/bin/pip install numpy markdown matplotlib pypdf schemdraw
```

System packages (not pip):

Need	Why	Notes
`kicad` 7.x, with `kicad-cli`	netlist export, SVG/PDF plots	schematics are generated, never hand-edited
KiCad stock symbol + footprint libraries	`/usr/share/kicad/{symbols,footprints}`	`hw/gen_footprints.py` derives from them
`libngspice.so.0`	the simulator, driven by ctypes	override the path with `NGSPICE_LIB=` if it lives elsewhere

Need	Why	Notes
`inkscape` , `imagemagick`	only for eyeballing rendered sheets	optional

On macOS (set up 8 October 2026, KiCad 10.0.7, ngspice 42): KiCad from kicad.org or `brew install --cask kicad`, and Chrome for the PDF build. **Use ngspice 42, not Homebrew's 47** — on 47, three blocks fail (`injector_turnoff` and `boost_converter` never finish, and `metering_unit_pwm` misses by 3 %), and the suite stalls at random. Build 42 from the release tarball (clang needs one warning disabled):

```
curl -fsSL -o ngspice-42.tar.gz \
  https://sourceforge.net/projects/ngspice/files/ng-spice-rework/old-releases/42/ngspice-42.tar
tar xzf ngspice-42.tar.gz && cd ngspice-42
./configure --prefix=$HOME/opt/ngspice-42 --with-ngshared --enable-xspice \
  --enable-cider --disable-debug --without-x CFLAGS=-O2 \
  CXXFLAGS="-O2 -Wno-invalid-specialization"
make -j8 && make install
```

Nothing lives at the Linux paths, so the scripts read these instead — put them in `~/.zshrc`:

```
export NGSPICE_LIB=$HOME/opt/ngspice-42/lib/libngspice.0.dylib
export SPICE_SCRIPTS=$HOME/opt/ngspice-42/share/ngspice/scripts   # else no spinit
export PATH="/Applications/KiCad/KiCad.app/Contents/MacOS:$PATH"
export KICAD_SYMBOL_DIR=$HOME/opt/kicad7-libs/kicad-symbols-7.0.11   # see below
export KICAD_FOOTPRINT_DIR=/Applications/KiCad/KiCad.app/Contents/SharedSupport/footprints
export CHROME="/Applications/Google Chrome.app/Contents/MacOS/Google Chrome"
```

Generate against KiCad 7.0.11's symbols, not KiCad 10's. The generators copy each library symbol into the sheet, so KiCad 10's library would change every symbol on every regenerated sheet (and `schlib` cannot read its tab-indented format anyway). With the 7.0.11 library an unchanged sheet regenerates identically, apart from UUIDs. KiCad 10 is used only for `kicad-cli`.

```
mkdir -p ~/opt/kicad7-libs && cd ~/opt/kicad7-libs
curl -fsSL https://gitlab.com/kicad/libraries/kicad-symbols/-/archive/7.0.11/kicad-symbols-7.0.1
```

The PDFs come out two pages shorter on this Chrome (326, not 328) with no source change — font metrics, not content. Don't commit a rebuild for that alone.

On Windows (set up 8 November 2026, KiCad 10.0.6), nothing beyond KiCad is installed. KiCad's bundled Python makes the venv (`…/KiCad/10.0/bin/python.exe -m venv .venv`, then `.venv/Scripts/python.exe`). It also runs `gen_pcb.py`, because it has `pcbnew`. Its `ngspice.dll` runs the simulations, and `pip install`

`ziglang` into the venv gives a C compiler (`.venv/Scripts/python.exe -m ziglang cc`) for the firmware, which has no `make` here: `run_sil.py` then uses the libraries already in `firmware/sim/build/`. Set:

```
export PATH="/c/Users/<you>/AppData/Local/Programs/KiCad/10.0/bin:$PATH"
export NGSPICE_LIB="C:/Users/<you>/AppData/Local/Programs/KiCad/10.0/bin/ngspice.dll"
export KICAD_SYMBOL_DIR="C:/Users/<you>/opt/kicad7-libs/kicad-symbols-7.0.11"
export PYTHONUTF8=1       # several files are UTF-8; Windows defaults to cp1252
```

Two traps: - **Write Python patches with `chr(92)` for a backslash.** The shell wrapper halves doubled backslashes before Python sees them, so a `"\\n"` written into a heredoc arrives as `"\n"`, a real newline. - **KiCad 10.0.6's schematic parity drops every pin on an auto-named net**, `Net-(...)` or `unconnected-(...)`, and reports "No corresponding pin found in schematic". It drops those pins and no others: 198 pads on the production board, 98 on the prototype. The netlist this same KiCad exports lists those pins on the right nets. 10.0.7 on macOS passed. Since 9 Nov, `check_pcb.py` checks exactly those reports against the exported netlist and passes a pad only if the schematic puts that pin on that same net. It prints how many it checked that way. Swapping two such pads' nets on a copy of the board still fails.

There is no Rust toolchain on that machine, so the simulator was not rebuilt.

Verify the whole project in one go:

```
set -o pipefail
.venv/bin/python sim/run_sim.py | tail -3        # expect 26/26 blocks pass
.venv/bin/python hw/check_netlist.py | tail -6   # expect "schematic connectivity OK", 0 open i
.venv/bin/python hw/assign_passives.py           # expect 253 packaged, 0 unresolved
.venv/bin/python docs/build_pdf.py | tail -1     # rebuilds docs/pdf/ECU-Bible.pdf
```

The schematics are generated. Edit `hw/gen_sheet_*.py`, then `.venv/bin/python hw/gen_sheet_<name>.py --force`. Without `--force` the generator refuses to overwrite a non-empty file. After adding or re-valuing any R or C, re-run `assign_passives.py` and then regenerate the sheets again so the footprints land.

Schematic UUIDs are derived, not random (since 15 Oct). A symbol's UUID comes from its sheet title and reference, and a sheet's from its name, so regenerating an unchanged sheet is byte-identical (`hw/test_schlib.py`). The PCB links every footprint to its symbol by these. Renaming a sheet, retitling it, or changing a reference re-links those parts. Before routing, re-run `gen_pcb.py`; after routing, use KiCad's Update PCB from Schematic with "re-link by reference". `check_pcb.py` fails on any stale link.

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3. Where things stand

Phase 2 (schematic capture) is complete and checked. Ten hierarchical sheets, root wired, project netlist verified. Phase 3 (part selection) is most of the way through. The PCB is prepared for hand routing (`hw/ecu25kva.kicad_pcb`, decisions log §13): every part placed by block, every rule set, with the housing and J1001 as placeholders until the old ECU is measured (brief Group 6). The firmware core runs on the PC against a simulated engine, with production logging (fault memory, events, counters, trip recorder) and all 16 scenarios passing (`firmware/`, decisions log §11–12). Since 20 October it also has a service link (decisions log §16, firmware guide §16): a bootloader, the end-of-line record with analog trims, an underspeed warning, and a laptop tool over USB or CAN channel 2. Only the chip layers are missing; the prototype is first flashed over SWD. **A simulator anyone can run** (`simulator/`, decisions log §17): `ecusim` is one Rust program that runs that firmware live or from scripts, traced in a browser, with firmware builds loadable from the page. It is built for a Windows user with nothing installed. **A prototype board is routed and fab-ready** (`prototype/`, decisions log §15): the full production circuit with an STM32G474 and screw terminals, 300 × 200 mm, 0 DRC errors, nothing unconnected. Fab files in `prototype/fab/`, bring-up guide `prototype/docs/bring-up.md`.

Gate	State
Simulation blocks	**26 of 26 pass**
Netlist connectivity	**passes**, with **0** open items (Bosch pinout, 8 Nov)
Footprints	**404 of 405** components
Chip R and C packaged	**253 of 253**

Verified part numbers are in `docs/bom_requirements.md` under "Chosen parts" — every one against a datasheet that was actually retrieved and read. That now covers the MCU, regulators, supervisor, gate drivers, current-sense and difference amplifiers, comparator, the boost controller, all sixteen power FETs, the small-signal FETs, 27 diodes, the gate-rail transistors, the buck inductor, and the packaging of every chip resistor and capacitor.

Design decisions taken with the owner are in `docs/decisions-log.md` — the four approved findings, the discrete-input redesign and the sensor-supply redesign, with the options and reasoning for each.

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4. What is open, in the order I would do it

4.1 Three approved redesigns: all three built

1. **POWER-PATH — built 8 October 2026.** Pin 04 now feeds `VBAT_INJ` and pin 06 feeds `VBAT_EGR`, each through a TVS, an SQM120P10-10m1L reverse P-FET and a Schottky (`power_path.cir`, and the netlist gate checks the split). Pin 21's L101, L102 and F101 are chosen for its new 2.95 A bound (`docs/bom_requirements.md` § POWER-PATH, with the thin margins stated). Left over: F101's pulse-cycling data. The two new rails' conducted emissions were closed 8 October with a filter on each feed (decisions log §10). It puts a firmware limit on hold chopping and EGR PWM: ≤ 25 kHz. The original item read: move the injector and EGR supply onto connector pins 04/06 (10 A fused each in the loom) with their own protection, leaving pin 21 feeding the electronics through the EMI filter and reverse FET. Today every high-current load runs through a path whose reverse FET was sized for 5 A and whose filter never stated a DC current. Blocks: L101, L102, F101. See `docs/bom_requirements.md` § POWER-PATH.
2. **SENSOR-PTC — built 8 October 2026.** The cut-off is a TPS48110-Q1 (chosen over the LTC4368 for AEC-Q100, decisions log §7). `sensor_rail.cir` is rewritten and the trackers' input peaks at 22.3 V. C211 and both land patterns were done 8 October. The original item read: one TPS7B4253-Q1 per sensor group, ADJ tracking 5V_MAIN, behind a 100 V-class overvoltage cut-off (an LTC4368-class controller was the candidate; **its datasheet was not retrieved yet — do that first**) with a 47 µF hold-up capacitor. The cut-off exists because the tracker's IN pin is rated 45 V and the rail reaches 73.3 V for 50 µs. Replaces F201–F203 and the whole SENSOR-PTC requirement. `sensor_rail.cir` needs rewriting with it, including the fault-containment check (one group shorted, the other two and the MCU unaffected).
3. **BOOST-SLOPE — built 8 October 2026.** VDD comes from a follower off the output (34.3–45.0 V), with the battery for start-up only. L401 is **150 µH**, Würth 7447709151 (decisions log §8 revised the approved 100 µH), and the sense resistor is 62 mΩ. Equation 9 holds by 1.12× at the worst corner, oscillator −20 % included. The original item read: feed the TPS40210's VDD from the clamped output side to steepen its internal ramp, and drop L401 to 100 µH (Würth 7447709101, Isat 3.1 A typ, AEC-Q200). Re-check TI Equation 9 at the cranking corner after the change; the current design is 1.83× over its ceiling there. The check that pins the finding is in `run_sim.py` under `boost_converter`.

4.2 Parts still unchosen

The three non-ceramic capacitors were chosen 8 October 2026 (decisions log §9): C313 is a KEMET T599 polymer tantalum with a 22 mΩ series resistor (R306) as its ESR floor. C211 is a Chemi-Con HXC 47

µF 63 V. C401 + C420 are two HXC 82 µF 80 V cans in series, balanced by the 2 × 11 kΩ bleed. What is left:

- **The trip-loop clamp** — the LOWV-CLAMP residual. It is the one input that still needs a continuous clamp, because its healthy band *is* the clamp level. It is also blocked on a connector pin.

4.3 Blocked on the machine or a supplier

- ~~**The power ground.**~~ Closed 8 Nov 2026: Bosch's terminal diagram puts it on pins 01/02, and the netlist check now requires it.
- `TRIP_LOOP` ~~**has no connector pin**~~ Removed 8 Nov 2026: no Bosch pin. The trip module opens the main relay's coil feed (spec §3 item 7).
- **At the machine (Bosch pinout):** pin 25's polarity, and what the GCU puts on 09 (analog) and 14.
- **The Bosch 94-way land pattern** — needs a distributor drawing for 1 928 405 192 / 194, code C. The board carries the OEM connector (spec §7 deviation 1 reversed), so this is the board's own header.
- ~~**U16, the alternator rectifier**~~ — closed 9 October by the owner. The board is designed for the suppressed (clamped, about 40 V) load dump, as most modern alternators are.
- **MC33816AE last-time-buy 06/08/2027** — needs a distributor, not another search. It only matters for the fallback path. (The L9781 item is closed: the owner supplied ST's data brief on 8 October, confirming its 80 V tank ceiling.)

4.4 The component still without a footprint

Parts	Why
J1001	the Bosch 94-way header: no public drawing. The harness mate is Bosch 1 928 405 063 (code C); the likely header is TE 936524-2 (154 = 60 + 94, 2.5 mm, restricted by TE). Ask the distributor, or buy and measure (memo 03 addendum). On the board as `Bosch_94_PLACEHOLDER` (6 large + 4 × 22, read off the old ECU); measure it from the old ECU (brief 6.3)
~~D806~~	the trip-loop clamp. Gone with the loop, 8 Nov 2026

`hw/gen_footprints.py` holds the project footprints. It derives two from stock by renumbering pads: a TO-277A and a SOT-23 zener, whose stock pad numbering would otherwise have reversed the part. It also **draws** four from the makers' recommended layouts (8 October): SMPD for D408–D414 (tab = cathode = pad 1), WE-PD 1050 for L201, TI DGX0019A for U203 (position 16 empty) and TI DDA0008J for U204–U206 (thermal pad 2.95 × 4.9 copper, 2.6 × 3.2 solder-mask-defined opening). They are written in KiCad 7 format. The installed KiCad 10 library is newer than KiCad 7 can open, so the script skips re-deriving the two stock copies from it. TI's optional thermal vias under the DDA pad are left to layout.

4.5 Then

Firmware on the S32K148 evaluation board, with an Arduino generating the engine signals (the core is already proven on the PC, §11 of the decisions log). Routing the PCB by hand (`docs/layout-guide.md`), once Group 6's measurements replace the outline and J1001 placeholders. Rebuild and check the board with the commands in the README ("Preparing the PCB"). The generator needs KiCad 10's own Python; one footprint KiCad 10 dropped (the TPS40210's HVSSOP-10) is kept from KiCad 7.0.11 in `hw/lib/kicad7.pretty` .

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5. Traps worth knowing before editing

These each cost a debugging session already:

- **KiCad's schematic-parity check doesn't notice a stale footprint link:** it falls back to matching by reference. `check_pcb.py` compares the paths itself.
- **kicad-cli 10 silently ignores a custom-rules file it can't parse.** Every number needs units (`0mm`). `check_pcb.py` confirms the file is the generated one.
- **A wire endpoint landing mid-span does not connect** in KiCad 7, even with a junction in the file. The page plots correctly and the netlist says "unconnected". Use `schlib.Rail` , which splits every segment at its taps.
- **A label's justification is read in its own rotated frame.** At 180°, "left" puts the text back across its own wire. Both label types are fixed in `schlib` ; keep it that way.
- **Pin numbers are not pad numbers.** A SOT-23 FET symbol (G-S-D) on a DPAK footprint puts the source on the tab. Power FETs use the G-D-S symbols; BJTs in SOT-23 are B-E-C; BZX84 zeners and TO-277A diodes need the renumbered footprints from `gen_footprints.py` .
- **Auto-generated net names move.** `Net-(R407-Pad2)` changes the moment a part is added ahead of it, and `assign_passives.py` keys voltages on net names — so give any node whose voltage matters a real label.
- `--force` , or the generator keeps the existing sheet.
- **The ngspice version matters.** The suite passes on ngspice 42 and fails on 47 (see §2). Upgrading the simulator is a change that needs the whole suite re-run, not a routine update.
- **kicad-cli's DRC report stops listing unconnected items at about 500.** A test that reads that list for one net passes blind on a badly broken board. Read connectivity in pcbnew instead (`gen_board.py clusters`).

- **pcbnew SWIG objects never match with `is`.** Each access hands back a new wrapper; compare `m_Uuid`. `f is not fp` silently let a new part collide with its own courtyard.
- **A symbol added ahead of others on a sheet renumbers them,** and the symbol UUIDs follow the references. On a routed prototype board, run `gen_board.py sync`: it matches footprints to parts by value and pad nets, renames them, refreshes their fields and places only what is new. A circuit change that renames nets or moves pins goes through `migrate` first (`BOSCH_MIGRATE` is the worked example). Then come `sync`, `finish`, `gnd_stitch` (ground vias for the new parts) and `tidy`. To move an MCU pin, `rip` its nets before `migrate` refits the pads.
- **The router walls in a pin between two busy neighbours.** Bosch's three new inputs on PE7–PE9 (bottom row, the CAN corridor) blocked PE8 and three CAN lines through two passes and a rip-up. On PD8–PD10, free pins facing J1007, they routed in one pass. Pick a pin facing its circuit before rip-up rounds.
- **The simulator's `EcuIo` copies `ecu_io` by hand.** Change the struct or an entry point, bump `ECU_ABI_VERSION` (`ecu.h`) and `ABI_VERSION` (`simulator/crates/sim/src/fw.rs`); `io_size_matches` catches a missed field.
- **A dynamic library loaded twice from one path is the first one again.** `Firmware::load` copies every upload to a fresh name. The C core's state is global, so a firmware is one ECU: a `Sim` claims it, and a second waits.
- **Simulation models hide things.** An idealised zener hid both the leakage that started LOWV-CLAMP and an 84 mV loading error that a passing check had pinned as the expected value. When a model's part gets chosen, re-fit the model to the datasheet.

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6. Repository map

Path	What it is
`hw/gen_sheet_*.py`	the ten sheet generators — the schematic source of truth
`hw/schlib.py`	drawing primitives, pin geometry, the chosen-parts table
`hw/assign_passives.py`	packages every chip R and C from the voltage across it
`hw/gen_symbols.py`, `hw/gen_footprints.py`	project symbols and renumbered footprints
`hw/check_netlist.py`	the connectivity gate: shadowed nets, decoupling, rails, open items
`sim/blocks/*.cir`	25 self-checking SPICE blocks
`sim/run_sim.py`	runs them and asserts every claim as arithmetic
`simulator/`	`ecusim`: the firmware against the engine model, in a browser (guide `simulator/README.md`)
`docs/bom_requirements.md`	requirements, chosen parts, and the open design items
`docs/status.md`	the long-form running record

Path	What it is
`docs/decisions-log.md`	decisions put to the owner, with options and reasoning
`docs/pinmap.md`	pin assignments, including the inverted discrete inputs
`docs/research/`	twelve research memos
`refs/`	OEM wiring diagram, machine photographs, extracted pinout, field evidence

`README.md` is stale in its counts (it said 163 pages, 17 blocks and nine memos; it is now 328 pages, 24 blocks and twelve memos, and it has been corrected). Worth a pass.